RNA-seq project:

Differences in pathogen response under varied environmental conditions

Background:

When environments change, the change can occur faster than normal evolutionary adaptive processes can act but plasticity in response to these rapid changes could allow organisms to withstand local conditions. Marine invertebrates have a large capacity for plasticity and exhibit it at almost any life history stage in response to both biotic and abiotic conditions. Among biotic factors that could influence them are microbes in the seawater. Seawater microbial communities, themselves, are susceptible to environmental shifts, with changes in temperature altering community composition and pathogen prevalence and spread. Because elevated temperature can also be a stressor, we are examining how *Strongylocentrotus purpuratus* larvae raised in different temperatures and microbial communities respond to infection.

These samples were collected in Spring 2024. Purple sea urchin larvae were raised in different environmental conditions, combinations of low or high microbial richness (LMR or HMR) and ambient or elevated temperature (14 °C or 18 °C). The larvae were infected with *Vibrio diazotrophicus* at 6 days post fertilization (6 dpf or 0 hours post infection [hpi]) and samples were collected 24 hours later (24 hpi). Samples were sent for RNA-seq to examine how the larval immune response differs between these environmental treatments.

Unique challenges:

- The *S. purpuratus* genome has had multiple genome versions and has changed the naming scheme for unconfirmed gene homologues between versions. Older versions used SPU IDs while the current version uses LOC IDs.
- The echinoderm community currently avoids naming genes until homologues are confirmed which means a lot of genes only have LOC IDs. This is also a recent change to naming conventions, which means genes have lots of synonyms, including possible homologues and previous SPU IDs.
- Both of the above can make comparison to previously generated reference lists and to other literature challenging.

Steps leading up to this point:

- Aliquots of 5000 larvae were collected and flash frozen in liquid nitrogen. RNA was extracted using a TRIzol:Choloroform phase separation protocol.
- RNA was sequenced using NovaSeq PE150.
- Sequences were filtered using FastQC and aligned to the genome of *S. purpuratus* using Hisat2.
- The aligned reads were analyzed with DESeq2 in R to determine the significance and fold change of differentially expressed genes.
- These data sheets have lists of genes and their adjusted p-values and $\log_2$ fold change values under different comparisons. The four lists were created by comparing a given condition at 24 hpi to all the other conditions.

- Genes that have positive number fold changes in these lists have higher expression at 24 hpi in this treatment group compared to the other groups.
- Genes that have negative number fold changes in these lists had higher expression at 0 hpi in this treatment group compared to the other groups.

Analysis steps:

1. Visit GitHub and find the script you'll be referencing. We'll be using the "RNA-seq_DEGclasses" script in the "CURE_scripts" repository. https://github.com/amyltan
2. Install any R packages that you're missing.
3. Download the two reference files and four data sets from Google Drive.

refseqLocus_spu_IDmapping.txt	Old and new gene names
Tu_Ontology_SuppTableS2.csv	Gene ontology list from Tu et al. 2012
res14lmr24-MM-DEG-Oct25.csv	Results for the larvae raised in 14 ºC LMR
res18lmr24-MM-DEG-Oct25.csv	Results for the larvae raised in 18 ºC LMR
res14hmr24-MM-DEG-Oct25.csv	Results for the larvae raised in 14 ºC HMR
res18hmr24-MM-DEG-Oct25.csv	Results for the larvae raised in 18 ºC HMR

4. Load the reference files.
5. Based on the list of gene classes, select one class to investigate today. (Exclude: Immunity, Biomineralization, Histone, Oogenesis, and ZNF.)
6. Run the script—making any necessary naming and coding adjustments as you go—for each of the four data sets.
7. Create a plot of the differential gene expression for each data set and answer the questions below.

Questions:

- Do any significantly different genes show up in multiple data sets? only in a single data set?
- Select 5 genes that show the greatest differential expression between conditions. What can you find out about the function of these genes?
- What might these differences tell us about how the larvae are responding to pathogens?
- If a database like Echinobase doesn't have much information about the gene, what other tools might be available for sussing out the possible identity and function of the gene?
- Can you think of any ways to show the gene expression results from all four lists in a more condensed format like a single plot?